CHAPTER - 2
REVIEW OF LITERATURE

2.1 INTRODUCTION

Writing a summary, synthesis, and/or critical analysis of the literature that is discovered as a consequence of a literature search is known as a literature review. It could serve as background information or context for a primary study endeavour.

A literature review is a procedure as well as an end result. It entails looking up information about your subject, becoming familiar with pertinent research, and identifying problems and gaps in the literature. Generally, you are not trying to read everything that has been published on the subject; rather, you are trying to find the major writers and points of contention that are pertinent to your issue.

2.1.1 Types of Review of Literature

There are numerous varieties of literature reviews, and each has a distinct methodology, analysis, and goal. Some of the more popular forms of literature reviews are listed here.

A literature review that stands alone can be considered a separate piece of writing. A review article and a class assignment are two examples.

Reviews of the literature can also be constituent pieces of longer works. A thesis or dissertation and the introduction to an academic journal article are two examples.

Systematic Review Medical trials, health care, and other fields where data and methods are crucial depend on systematic review. The systematic review finds and contrasts responses to health care-related questions by carefully reviewing and analyzing literature that satisfies a set of criteria. Meta-analysis and meta-synthesis may be included in the systematic review, which brings us to.

Quantitative or Qualitative Meta-analysis Review Meta-analytic Reviews might comprise a systematic review in its entirety or in part. Both distill and make sense of a substantial body of research in a thorough and complete manner. Reviewing quantitative research, the quantitative meta-analysis is impartial and incorporates statistical analysis. Reviewing qualitative research, the qualitative meta-analysis finds new themes or concepts and is subjective (evaluative, interpretive, or both).

Literature Review (see our Literature Review video) **or** A thesis or dissertation frequently includes a chapter on narrative review. It explains the relevant research that has already been done, how it influenced the thesis, and how the thesis fits within the body of field research.

Critical Review Similar to a literature review, a critical review entails a more thorough analysis of the literature in order to compare and assess various points of view.

Scoping Review Examining A review is frequently used at the start of a research proposal, dissertation, or article. It is carried out ahead of time and provides context for the research by drawing attention to gaps in the literature and outlining the reasons behind the impending study, which is detailed in the body of the article.

Conceptual Review organizes papers based on topics, categories, or concepts. It outlines the present "understanding" of the specified study topic, talks about how this understanding was arrived at, and looks for indications of whether a deeper understanding might be proposed. It offers a quick overview of the current state of this specific area of study.

State-of-the-Art Review Current Situation Periodically, an art review is undertaken with an emphasis on the latest studies. It outlines what is currently understood, agreed upon, or known about the study issue and points up areas in which differences still exist.

2.1.2 Methodology of Literature Review

As explained in Templier and Paré (2015), there are six generic steps involved in conducting a review article:

Formulating the research question(s) and objective(s),

Searching the extant literature,

Screening for inclusion

Assessing the quality of primary studies,

Extracting data, and

Analyzing data.

2.1.3 Systematic Literature Review

A systematic literature review (SLR) identifies, selects and critically appraises research in order to answer a clearly formulated question (Dewey, A. & Drahota, A. 2016). The systematic review should follow a clearly defined protocol or plan where the criteria is clearly stated before the review is conducted. It is a comprehensive, transparent search conducted over multiple databases and grey literature that can be replicated and reproduced by other researchers. It involves planning a well thought out search strategy which has a specific focus or answers a defined question. The review identifies the type of information searched, critiqued and reported within known timeframes. The search terms, search strategies (including database names, platforms, dates of search) and limits all need to be included in the review.

Pittway (2008) outlines seven key principles behind systematic literature reviews

- Transparency
- Clarity
- Integration
- Focus
- Equality
- Accessibility
- Coverage

2.1.4 Literature Reviewed for Study

There have been a lot of studies conducted all over the world using Bibliometrics and Scientometric techniques on various types of diseases. The selected reviews for the study have been summarized based on various facets of bibliometrics study. The current literature review is based on my topic **"A Scientometric Study of Research Output on Coronavirus"** The few of them are as follows:

2.1.5 Reviews Statistics

In this review of literature 80 (see Figure 2.1) reviews are taken from reputed journals articles.

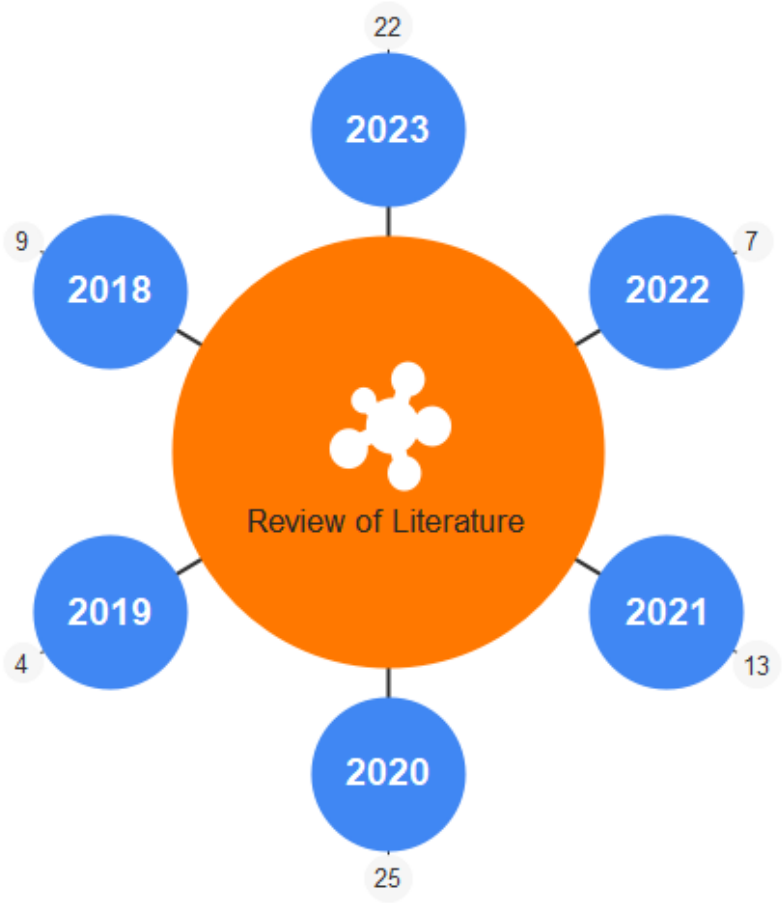


Figure 2.1 Statistics of Reviews

Bhowmick and Chakrabarty (2023) presented Scientometric study on COVID-19 has taken place to highlight the latest trend and situation of literature on COVID-19. The coronavirus spread around the world after the Wuhan pandemic, wreaking havoc on socioeconomic management, movement, livelihood, and health. The outcome of an effort to do a qualitative and quantitative analysis is this scientometric study. The number of research outputs is increasing by thousands as a result of this epidemic situation. The WOS database indexed 1234 articles of literature about COVID-19 and related topics between December 2019 and May 2021. To facilitate more contact, most publications took down their paywalls. A string was created in order to retrieve all secondary bibliographic data. A text file (.txt) has been extracted from the WOS database and utilized for mapping and visualization. A few statistical formulas can be used to calculate the collaboration coefficient (CC), author productivity, and annual growth rate (AGR). Applications for data representation are also used, including QGIS, MS Excel, and WOS viewers. The outcome causes the literature growth map to

continuously rise. Among the top ten journals and authors, "Medicine" and "Wang J" ranked highest. Even though author collaboration peaked at 0.997, more cooperation was still required. The most pertinent research output was created by China, India, and the United States. All subject areas, including engineering, social science, environmental science, and medical science, are close to COVID-19. The results of this study are supposed to help society and the government manage and create regulations that will prevent such natural disasters in the future.

Boubaker et al., (2023) have presented COVID-19 has posed unprecedented challenges to global finances because of its unparalleled global scope, with both concomitant shocks as well as the likely altering of risk assessments and forecasts for the foreseeable future. Since a lot of research has been done on the effects of COVID-19 on financial markets and institutions, we comprehensively summarize it here. After a thorough search procedure, we extract and analyze 818 articles. Using bibliometric techniques, we investigate the patterns among different research participants in the field. We delineate four primary themes and the intellectual framework of the field's research using multi-dimensional scaling. In the brief three years after the launch of COVID-19, we also chart the changes and advancements in research. A thorough content analysis is used to suggest several future study directions.

Cano-Marin et al., (2023) stated that Interest in healthcare has grown significantly worldwide, especially since the COVID-19 outbreak. Because digitalization has made it possible for people to communicate on social networks via Twitter and other platforms, user interactions have been collected over time, which has led to the spread of fake news. The objective of this study is to assess, categorize, and analyze Twitter analytics' predictive capabilities in the healthcare industry. It also seeks to uncover hidden knowledge insights and separate them from related rumors and false information. Thus, in order to find and evaluate the current scholarly research and applications in Twitter for healthcare prediction, a systematic literature review (SLR) is conducted. The identification of public health emergencies and mental health disorders are the two most significant predictive applications. The majority of research on fake news and misinformation has focused on COVID-19; this study adds something useful to the usage of unstructured data from Twitter and highlights the significance of this content when it comes to healthcare. Since it is crucial to assess

the accuracy of the information given on Twitter, it is vital to concentrate on the advancements provided by these data as a prediction tool in healthcare.

Cataldo et al., (2023) examined Since the outbreak of COVID-19, researchers and clinicians have published scientific articles on the SARS-CoV-2 virus and its medical, organizational, financial, and psychological implications. Nonetheless, a number of consequences have been noted in the post-lockdown situation. In order to represent the state of the art with relation to the psychological and emotional impacts of the pandemic following the lockdown, we used a scientometric–bibliometric strategy in this study. We located 791 publications in Scopus that were then examined with CiteSpace. Eight primary clusters were produced by the document co-citation analysis (DCA) algorithm, each of which represented a focal point of study. One significant cluster, cluster no. 1, is specifically concerned with the long-term consequences of the COVID-19 pandemic and people's capacity to build resilience and adaptive coping strategies. The findings enable us to more clearly define the research fields that have been studied and the areas that still require more inquiry.

Chen et al., (2023a) presented Coronavirus disease has caused city blockades, making people spend longer in residential areas than ever before. The epidemic can be directly controlled by home planning and design, which has an impact on human health and well-being. To help the states prepare for the long-term effects of COVID-19, researchers from all around the world have been working hard to investigate the connections between residential planning and design and the virus. The scientometric analysis of the literature on "Effects of COVID-19 on residential planning and design (ECRPD)" published in Web of Science and Scopus from 2019 to October 2022 is the basis of this study, which uses Bibliometrix as its database. In order to support and advise urban planners, builders, and policy makers, this study aims to comprehensively present the scientific knowledge of ECRPD research through general characteristics' analysis, citation analysis, and horizontal conceptual structure analysis. It also attempts to summarize how residential planning and design responds to COVID-19. The findings show that ECRPD research is expanding rapidly and that its scientific productivity has skyrocketed. Three aspects—the residential environment, residential building and planning space, and residential traffic and community management—are indicative of the primary consequences and feedback.

In general, research on ECRPD has surpassed the boundaries of planning and architecture. The most attention has been focused on environmental and energy issues, notwithstanding the paucity of empirical study on residential construction space. The development of bioclimatic buildings to adapt to environmental changes, the promotion of cross-disciplinary and multi-field collaboration, the application of new technologies and methods for traditional disciplines, and the strengthening of practical research in residential building and planning are all necessary to fully address the multiple negative aspects of COVID-19. These measures will ensure that a resilient and sustainable living environment is created in the post-pandemic era.

Chen et al., (2023b) has examined COVID-19 has a significant impact on dental medicine. The current study uses the visual mapping method to provide an overview of COVID-19 dental research. Methods: We examined the publications in the Web of Science core collection under the category "Dentistry Oral Surgery Medicine." In order to find dental publications that discuss COVID-19 or SARS-CoV-2, we performed an advanced search on June 10, 2022, using the items TS = ("Novel coronavirus 2019" or "COVID 19" or "Coronavirus disease 2019" or "2019-nCOV" or "SARS-CoV-2" or "coronavirus-2) and WC = ("Dentistry Oral Surgery medicine"). With VOSviewer and Microsoft Excel 2010, the contributions of writers, journals, institutions, and nations were detailed. VOSviewer and CiteSpace were used to visualize the co-occurring analysis and references analysis of the keywords. Findings: Between 2020 and 2022, 1,732 papers were found. Three significant contributions to this field were Brazil, the United States, and the United Kingdom. With 55 articles in this discipline, University of São Paulo (Brazil) came in first. With 19 publications, Hercilio Martelli Junior from Universidade Jose do Rosario Vellano in Brazil was the most prolific writer. The two most productive journals were Oral Diseases and British Dental Journal. Dental practice and infection control, COVID-19 oral manifestation, dental education and online learning, teledentistry, and mental health issues were the main subjects covered. In summary, there has been a notable increase in the growth rate of publications pertaining to dentistry research on COVID-19. In 2020, the focus of study switched from "Oral manifestation related to COVID-19, dental practice and infection control" to three key areas for future research: "Dental education and online learning, teledentistry, mental health problems."

Damaševičius and Zailskaitė-Jakštė (2023) have studied the coronavirus disease 2019 (COVID-19) pandemic has greatly impacted society and academic life and research practices. The goal of this study is to determine whether the COVID-19 pandemic's worldwide emergency has an effect on international researcher collaboration. To understand how scientists interacted both within and outside of their own country, the authors examine research collaborations conducted prior to and following the start of the COVID-19 epidemic. Design/methodology/approach: To understand how scientists interacted both within and outside of their own countries, the authors examine research collaborations conducted prior to and following the start of the COVID-19 epidemic. The researchers gathered a dataset of research articles from 11 different countries (Austria, Denmark, Greece, Indonesia, Iran, Ireland, Korea (South), Mexico, Pakistan, Romania, and Vietnam) that were published in journals within the field of business and economics and indexed in the WoS Core Collection database. 14,824 publication records in total were taken into account for the literature review. The scientometric analysis of these papers was presented in this study utilizing statistical, factor analysis, network analysis, and bibliometric techniques. The Hofstede model of cultural dimensions is used to assess and interpret the findings. The study's findings show that research management should focus their efforts appropriately to enhance researcher collaboration amidst the COVID-19 epidemic and mitigate its adverse effects in the future. Results - Our study's findings demonstrate that the Hofstede model's definition of uncertainty avoidance as a cultural component has a major impact on the characteristics of research collaboration networks in the business and economics domains. The focus of uncertainty avoidance is on how societies adjust to change and deal with uncertainty; nevertheless, the global COVID-19 epidemic brought about a great deal of uncertainty and change at all societal levels. Research constraints and ramifications: This study focused solely on business economics and only looked at 14,824 research outputs that were indexed in the WoS Core Collection database between 2019 and November 15, 2020. Therefore, papers that have been published in sources and channels that are not covered by WoS are not included in this analysis. Just 11 nations' worth of publications, or a tiny portion of the world's research output, have been examined by the writers. Furthermore, there are other methods for examining cultural traits at the national level outside the Hofstede's cultural dimensions model. Practical implications: Research management will be able to better focus their resources in light

of the study's findings to enhance researcher cooperation during the COVID-19 epidemic, which is still underway, and to mitigate its detrimental effects in the years ahead. Originality/value: Taking into account the COVID-19 pandemic's worldwide effects and social unrest, this study is important because it is the first of its kind to identify changes in the features of research collaboration networks of 11 diverse (different in terms of geographic distribution and cultural differences according to Hofstede's cultural dimensions model) countries between 2019 (the year before COVID-19) and 2020 (the year of COVID-19).

Dash (2023) Presented COVID-19 is the latest pandemic form and, as usual, it also opened gates for new avenues in consumer behavior research. This study focuses on customers' panic buying as a result of the ongoing pandemic. In order to examine the research published in the recent three years (2020–2022), this study uses a bibliometric technique (clearly, no research before 2020). Based on the Web of Science database, 79 studies are listed. Initially, a broad examination is carried out to comprehend the present status of research by charting nations, writers, journals, references, establishments, and so on. It offers the correlation or link between the factors stated and the studies cited. Second, deep digging and targeted keyword analysis are used to provide insights and recommendations. The results show that panic buying did occur during COVID-19, particularly in the first year, and that several factors contributed to this behavior. The second year saw a significant decline. In the event of another pandemic, this study offers guidelines for practitioners on how to handle panic buying.

Gomis et al., (2023) The present study aims to analyze to significant increase in the literature on learning resources in Higher Education (HE) but very limited evidence of studies that have taken a global overview of the context, range, and emerging trends from the previous research. In order to provide a worldwide overview and research trends under the topic of learning resources in HE, this study intends to do a Scientometric analysis of research papers. The dataset, consisting of 4489 scientific articles published between 1970 and 2022, was sourced from the Web of Science database. Co-authorship analysis for authors, organizations, and nations, as well as co-occurrence analysis for keywords from the VOSviewer software, were used to obtain network maps and important data. According to the survey, the United States

of America made a substantial contribution to research, with Salamin, N. from the University of Lausanne being identified as the most frequently published author. The University of Illinois in the United States has made the most contribution to research publications. The most popular research trends and hotspots were cloud computing, e-learning, education, academic libraries, and learning resources. The study's most important conclusion, however, is that genuine collaboration is required within the research theme and offers suggestions for how to strengthen partnerships in order to improve HE learning materials. This study might be the first to analyze learning resources in higher education using scientometrics. The present study provides significant understanding of the HE learning resource landscape to scholars, academic institutions, researchers, policymakers, and pedagogical statutory bodies. It also identifies areas for future research, collaborations, and policy development based on the study's critical findings.

Guo et al., (2023) The present study aims to analyze the evolution of research on children and adolescents' mental health issues during COVID-19 pandemic and discuss research hotspots and cutting-edge developments. Methods: Citespace, VOSviewer, a bibliometric visualization mapping program, was used to evaluate the literature that was retrieved from the Web of Science Core Collection as of June 28, 2022. Findings: 5,594 of the 6,039 pertinent publications that were located were included in the analysis. Since 2020, there has been an increase in the quantity of literature; analysis was done on journal articles, institutions, and countries. The co-citation analysis reveals that there are less systematic reviews that employ critical thinking for review and more research papers among the highly referenced articles. The most representative categories in the cluster analysis were life transition and mental health. The keywords' timeline view reveals that the three biggest clusters—online learning (#0), public health (#1), and mental health (#2)—are changing over time. In summary, this research contributed to the analysis of children's and teenagers' mental health during the COVID-19 pandemic and revealed significant trends and deficiencies. These findings will serve as valuable references for future research and decision-making, as well as technical advice for systematic reviews.

Gupta et al., (2023) Open banking is an emerging and evolving concept, it brings both opportunities and challenges to banks. The topic has received little attention

from academics, despite its potential to revolutionize the financial services industry and provide customers with better value. This paper summarizes the body of knowledge regarding open banking and its growing significance. It reviews the body of literature in this area using scientometric technique and underlying statistical analysis tools. For this investigation, data from 415 documents were examined closely. We conducted a literature analysis in order to provide the research community with an overview of the following topics: research trends, major themes, associated keywords, and potential future research avenues. Aspects of the authors' analysis, source analysis, author cooperation network across national boundaries, keyword analysis, and topic modeling were disclosed by the scientometric study. This work adds to the body of literature by compiling and evaluating the body of open banking research that has already been done.

Hamidi et al., (2023) have focused During the COVID-19 pandemic, modern science demonstrated its ability to respond well to the health crisis by publishing useful and reliable information. The number of psychological articles in this sector has also increased as a result of this illness. However, social science research has been disregarded in favor of medical issues in the majority of scientometric investigations. In order to close this research gap, we examined the psychological literature on COVID-19 in order to shed light on the viewpoint, areas of study, and global partnerships. Web of Science data was gathered, and Citespace and Bibliometrix (Biblioshiny) were used for analysis. Following a description of the documents' overall performance, networks representing keyword co-occurrence and co-authorship were shown. Document co-citation network drawing resulted in the formation of fifteen major clusters. The findings suggest that researchers were interested in the following areas: anxiety, mental health, delirium, loneliness, and suicide. Considering the unique circumstances that COVID-19 brought about for human cultures, psychological research is arguably one of the most significant topics in the medical industry. Psychology researchers can use the study's findings to determine future research partners and areas of overlap in the COVID-19 field.

He et al., (2023) has explored the antiphospholipid syndrome is an autoimmune disorder characterized by recurrent vascular thrombosis and pregnancy losses in the presence of persistently positive antiphospholipid antibodies. Academic productivity

and influence can be analyzed using altmetric and bibliometric techniques. As of right now, the antiphospholipid syndrome cannot be adequately assessed using the aforementioned analyses. Through the use of bibliometric and altmetric analysis, this study examined the volume and caliber of research conducted on antiphospholipid syndrome and identified the traits of global productivity in this area. Techniques From January 1, 2011, to December 31, 2021, the terms "APS," "antiphospholipid syndrome," "antiphospholipid-antibody syndrome," and "Hughes syndrome" were searched on the Web of Science Core Collection. Reviews and original pieces were chosen. The filters are set to "English." Findings and conclusion Twenty of the 68 nations from which a total of 1818 articles were collected fit the definition of major active countries. 1341 papers (73.48%) were contributed by high-income nations. Throughout the ten years, there was a significant annual rise in the number of publications (P<0.001). The nation with the highest productivity was the USA (253, 13.91%). Serbia topped the list when population was taken into account. Serbia was ranked first by the gross domestic product analysis. By using keyword analyses, the most popular terms, such thrombosis and antiphospholipid antibodies, were displayed. Anticoagulant and antithrombotic therapy were identified as research hotspots by a content analysis. Average citations were shown to be significantly correlated with both Mendeley readers count (P<0.001) and altmetric attention scores (P=0.002). There was a substantial rise in the quantity of worldwide articles between 2011 and 2021. The majority of the papers were from wealthy nations. Since there was a generally consistent relationship between the bibliometric and altmetric analyses, they might support and enhance one another.

Heidari and Olivieri (2023) The present study offer a thorough scientometric analysis of the practice of bioclimatic retrofitting in commercial buildings, which is considered a crucial approach for mitigating energy consumption and addressing the challenges posed by climate change. Using a bibliometric search, a scientometric methodology, and a thorough qualitative analysis, the database of Scopus was utilized to obtain bibliographic information for around 400 published papers because it provides sophisticated tools for literature search and analysis. A range of quantitative parameters, including the number of intermediate citations, occurrences, average publication year, and general connection power, are employed in the scientometric approach. An analysis of research publishing trends from May 2022 to the end of

2008 has revealed an increasing trend in academic research on commercial buildings over this 15-year period. However, studies conducted in the context of commercial buildings on the subjects of bioclimatic design, retrofitting techniques, and green building practices show a more gradual incline, with an almost linear trend between 2016 and 2020. The study also offers qualitative viewpoints on the research setting. The research priorities in the field are outlined in the paper. These include thermal comfort in relation to energy management and climate control, ventilation systems, sustainable development as it relates to architectural designs and green buildings, strategies for retrofitting commercial buildings to increase energy efficiency, the relationship between energy conservation and atmospheric temperature, the use of solar energy, and the use of building energy simulation techniques.

Huang et al., (2023) Since the mid-2000s, breast cancer incidence among women has slowly increased at about 0.5% per year. Over the past thirty years, research has demonstrated that the BRCA gene plays a critical role in promoting the occurrence and progression of breast cancer. There is, however, a paucity of scientometric analysis on breast cancer associated with BRCA. In order to ascertain the present state of affairs and stay abreast of the hotspots, a scientometric study was carried out on certain scholarly papers gathered from the Web of Science (WoS). To gather related papers for our dataset, we conducted a search on the Web of Science Core Collection (WoSCC). The software programs Bibliometric, CiteSpace, VOSviewer, and HistCite were then utilized to perform visual assessments of the nations, organizations, journals, writers, landmark papers, and keywords related to this field of study. Ultimately, 1,310 review articles and 7,266 publications published between 2013 and 2022 were retrieved. The yearly production increased year after year until reaching a peak in 2021. In terms of collaboration, total citations, and published works, the United States of America took the lead. In this field of study, Breast Cancer Research and Treatment were the most popular journal. The University of Toronto's Narod SA produced the most publications. Ultimately, the terms "breast cancer" (n=1,778), "women" (n=1,369), "brca1" (n=1,276), "ovarian cancer" (n=1,259), "risk" (n=1,181), and "mutations" (n=929) emerged as the most frequently used keywords, revealing the hotspots within the BRCA domain of breast cancer research. The scientometric analysis showed the trend in the BRCA research field over the previous ten years. The

clinical trials of poly-adenosine diphosphate ribose polymerase inhibitors (PARPi) and their resistance mechanisms are the main focus of current research.

Khan et al., (2023) this study employed a novel approach by analyzing the self-healing asphalt literature based on scientometric analysis. Co-citations, co-occurrences, and knowledge mapping and analysis are the most challenging aspects of contemporary research. The required data for the analysis, which included keywords, abstracts, citations, and bibliographic details, was located by searching Scopus. The most popular research sites, the publications that were cited the most during the data assessment process, and the most significant writers in the subject of self-healing asphalt were examined, along with the relationships between them. The primary issues of utilizing self-repairing asphalt were also highlighted, in addition to its necessity. According to the keyword analysis, self-healing asphalt has been researched by researchers as a sustainable maintenance technique for crack repair in asphalt road pavements. The review of the literature revealed that two methods of self-healing asphalt are heating and encasing rejuvenating chemicals. While the heating method employs microwave radiation and induction heating, encapsulation uses rejuvenating capsules. In order to improve asphalt self-healing, researchers have also created hybrid asphalt self-healing techniques. In order to create collaborative projects and disseminate novel concepts and methods, academics may find it advantageous to utilize the quantitative evaluation of areas and researchers in addition to the scientific description of these fields.

Mare et al., (2023) have presents COVID-19 has changed the world since 2020, and the field of water specifically, boosting scientific productivity (in terms of published articles). In relation to four water variables—wastewater, renewable water resources, freshwater withdrawal, and access to improved and safe drinking water—this research examines the impact of COVID-19 on scientific production. The maps were constructed after the body of literature in the topic was examined, highlighting the close relationships between COVID-19 and variables connected to water. A total of 94 nations were taken into consideration and their clustering was assessed based on articles that compare COVID-19 to water. The last stage of the study reveals that, generally speaking, the majority of research on the issue of water was produced in nations with lower rates of COVID-19 infection but greater levels of development as

indicated by the human development index (HDI) and gross domestic product (GDP) per capita. The statistical analysis shows that the variables connected to water have positive coefficients and are extremely significant. This demonstrates that nations with higher water-related values have studied the connection to COVID-19 more thoroughly. Regarding COVID-19, freshwater and wastewater extraction had the most effects on scientific productivity. Given the development parameters, having access to clean drinking water becomes irrelevant.

Paeffgen (2023) has paper examined Since the start of the pandemic, organisations have recognised the importance of resilience strategies and they have been trying to enhance them for future crises. COVID-19 has had a significant impact on how organizations function and respond to current events. It is yet unknown what lessons this pandemic can teach us about unfavorable future occurrences. The purpose of this study is to pinpoint the most important problems and gaps in the literature pertaining to organizational resilience in the COVID-19 environment. In order to do this, a bibliometric literature review of 185 papers covering organizational resilience throughout the COVID-19 pandemic, from 2019 to the start of 2022, is being conducted for this research. Six primary issue clusters are identified in this paper: (1) emergency responses, tourism, and entrepreneurship; (2) disruptions, organizations, and the global environment; (3) employees and mediation; (4) firms and opportunities; (5) capacities and digitalization; and (6) leadership in uncertain times. This study's primary contribution is a current inventory of studies on organizational resilience during crises, especially those carried out during the COVID-19 pandemic. The COVID-19 pandemic and the timely outcomes of this paper are noteworthy because of their uniqueness. Given this, I think it is possible to create novel, ground-breaking new paths for resilience literature in the future.

Satish et al., (2023) South Africa reported a new variant of SARS-CoV-2, named 'Omicron,' to the World Health Organization on November 24, 2021. Scientometric analysis quantifies everything currently available written communication, along with its authors and citation analysis. The study's goal was to use the Scopus and Web of Science databases to analyze the research metrics of Omicron publications published between January 1965 and March 2020.All scientific research outputs with the word "Omicron" in the title were retrieved from the Web of Science (January 1965 to

March 2022) and Scopus (January 1896 to March 2022) databases on April 10, 2022. The data was analyzed according to the following criteria: year of publication; top authors; most popular journals; highly cited articles with citations; publication type; subject classification; frequently used keywords; publication distribution geographically; language; and additional institutions that contributed. 1229 research outputs were for WoS, and 655 were from Scopus. While Scopus grew from 1 in 1896 to 356 in 2022, the number of articles for WoS increased from 22 in 1965 to 265 in 2022. Yuen KY, the author, was found to have contributed the most to both databases; the majority of the articles were original, with the majority published in English; the most productive country was the United States of America; the journals that were most frequently cited were the Lancet and the Journal of Virology and Nature; and the majority of the articles were published in English. The majority of the articles by WoS and Scopus, respectively, were on the topics of Chemistry and Medicine. According to WoS and Scopus, the most productive universities were the "University of California Systems" and the "University of Hong Kong." In the late 1700s, the word "Omicron" initially surfaced in the field of chemistry. The publication's growth pattern was irregular. In the upcoming days, researchers will find the aforementioned research metrics useful.

Singhania and Chadha (2023) the growing relevance of sustainability reporting (SR) has dramatically surged advocacy and interest among both academicians and practitioners. However, few studies have attempted to holistically encapsulate global research on sustainability reporting. In order to map the intellectual structure of the subject of sustainability reporting in detail, the current study used scientometric analysis on 1434 papers that were taken from the Web of Science database and published between 1992 and 2022. CiteSpace software was utilized to create domain visualizations that included co-citation and cluster analysis, as well as networks of co-authorship, keywords, subject categories, institutions, and nations involved in publishing on SR. The results showed that developed countries have made the majority of the major contributions to SR research, highlighting the need for additional study in the context of underdeveloped and emerging nations. It was discovered that while the SR field lacked international collaboration, its research sub-communities were cohesive. The subject areas of business, management, environmental studies, green and sustainable science and technology, environmental

sciences, and business finance were the main focus of previous study in the SR research domain. Pioneer works in the subject were detailed through content analysis of highly co-cited articles and examination of the authors who were most frequently cited. The main subjects covered by the corpus of literature were found through groups of documents and keywords that co-cited each other. Future research priorities include examining the relationship between the circular economy and social responsibility (SR), the role of digital technologies such as artificial intelligence, blockchain, and social media in SR, focusing on the MSME sector, requiring mandatory reporting, evaluating the true impact of SR on investor sentiment and financial analysts' valuations, providing assurance, standardizing research, addressing materiality concerns, and comprehending SR's niche themes, including monothematic reporting. The study's implications for businesses, society, academics, and legislators were looked at.

Verma et al., (2023) A scientometric analysis of funded scientometric and bibliometric research from 2011 to 2021 is presented in this paper. We examined 1040 methodologically relevant publications (MRP) and 2810 domain-related papers (DRP) using Web of Science. The study has looked at a number of common scientometric metrics, such as the h-index, the average citations per paper (ACCP), the doubling time (DT), the activity index (AI), and the relative growth rate (RGR). The parameters linked to the most prolific funding agencies, patterns of foreign country collaboration, top journals, most productive institutions, field trends, and emerging subjects are all thoroughly analyzed in this study. For the first time, we examine scientometric and bibliometric research funding according to study type in this publication.

Yalley et al., (2023) have analyzed the COVID-19 pandemic has generated worldwide research efforts to provide knowledge about the disease. However, not much is known about Ghana's role in this important body of knowledge creation. The purpose of this scientometric analysis was to comprehend Ghana's COVID-19 research output in the African context and to pinpoint possible future possibilities. In December 2022, the study pulled pertinent material from the Scopus database that covered the years 2019 to 2022. A number of established indexes, such as collaboration patterns, productive institutions, citation patterns, and important

research funders, were used to evaluate the retrieved data. Ghana published 1112 times in total, ranking eighth in Africa. South Africa was the principal African collaborator with Ghana, while the United States and the United Kingdom were the key international partners. There is a significant gender disparity in Ghanaian research production; of the top 21 most productive writers, 85.7% were men and 14.3% were women. Africa's total research production is limited by the lack of intra-continental research cooperation, despite Ghana's modest contributions to the worldwide COVID-19 research output. Our research shows how important it is for the Ghanaian government to fund research at the highest priority and remove obstacles that prevent women from participating as productive researchers.

Apitz and Fontoura (2022) It is unclear if research published in primary care journals aligns with the broad spectrum of problems managed in primary care practice. In order to identify research gaps, the purpose of this study was to analyze publishing trends regarding the burden of medical conditions reported in primary care journals and compare these findings with the burden of problems observed in clinical practice. A total of 9956 articles from primary care journals indexed in MEDLINE were analyzed using scientometric methods. Using keyword research, a relations map was constructed. A primary care database and a literature search were utilized to determine current issues and justifications for seeing a family doctor. Rankings and frequency of circumstances and research output were contrasted. Five clusters of publication patterns were detected using keyword analysis: oncology and health management; mental illnesses; infections; cardiovascular ailments and conditions associated with bad lifestyles. The domains of orthopaedics, endocrinology/metabolism, gastroenterology/hepatology, dermatology, ophthalmology, and the respiratory system have the largest gaps when compared to publications with clinical concerns. More specific possible study subjects, including burnout, osteoporosis, osteoarthritis, COVID-19, chronic pain, sleeplessness, and prescription antibiotics, were discovered through the relations map. The distribution of issues encountered in clinical practice and the rationale behind seeing a family doctor differs from the distribution of publications in primary care journals. One way to find areas where primary care research is lacking is to use scientometric methods to analyze publication trends and compare them to common concerns.

Atlasi et al., (2022) This paper tags the Many studies have suggested herbal medicines as alternatives or adjuvants to modern drugs for COVID19. A scientific synopsis of this subject can be obtained from their scientometric assessments. Materials and procedures: We searched Scopus and Web of Science (WOS) for articles published up until October 26, 2020, about the use of herbal medications in COVID-19. The collected data were subjected to statistical analysis using VOSviewer 1.6.15, ScientoPy 2.0.3, Gephi 0.9.2, and SPSS 15 for document type, subject area, top journal, citation number, and authors' collaboration network. Results: 378 and 849 records, respectively, from WOS and Scopus, remained for quantity analysis after screening the 3185 retrieved records. The two primary categories of articles in both databases were original and review articles. The WOS and Scopus databases' top subject areas were, respectively, drugs and medicine. China, the United States, and India were the three most productive nations in the field. In both databases, a practice guideline was the most cited article. The top journals were "Chinese Traditional and Herbal Drugs" in Scopus and "Journal of Biomolecular Structure Dynamics" in WOS. "COVID-19" and "Traditional Chinese Medicine" were among the top keywords. US writers collaborated with other writers at the highest rate. Conclusions: This study offers a glimpse into the amount and nature of scholarly publications that have been released in the last several months about the relationship between COVID-19 and herbal remedies. Our results assist researchers in identifying current gaps in knowledge, locating active authors and scientific institutions with whom to work, and leveraging their expertise to generate new discoveries in the future.

Mahdavi et al., (2022) have highlighted During COVID-19 pandemic, people have been reluctant to visit dentist due to the fear of infection. Dentists have experienced extreme unrest as well. One important tactic for controlling people's oral health is teledentistry. This study looked into teledental care during the COVID-19 pandemic around the globe in an effort to gather insights from various experiences. Materials and procedures: Using scientometric analysis, the Web of Science database was searched using keywords like COVID-19, telemedicine, teledentistry, and MeSH-derived synonyms. After extracting 94 articles, 15 publications about teledentistry during COVID-19 were included in the study after taking into account the inclusion and exclusion criteria. Results: To identify authors, organizations, and nations, as well as to examine networks of cooperation and collaboration in this field, scientometric

analysis and illustration of the retrieved papers were carried out. During COVID-19, telemedicine was researched in China, the US, Japan, Italy, and the UK, among other nations. WeChat, BigWord telephone translator, iGAM, WhatsApp, Attend Anywhere (AA), and DeRS were some of the software types utilized to facilitate communication between dentists and patients. Proforma and Mouthwatch TeleDent were the programs utilized to save the data. Automated algorithms were used for data processing and decision tree methods. In conclusion, teledentistry holds promise for enhancing the delivery of dental care. Providing such services requires meticulous planning and a fundamental assessment, especially in emerging nations. Moreover, further comprehensive research is necessary to address current and potential hazards and limitations related to teledental therapy while optimizing its benefits.

Mayta-Tovalino et al., (2022) have Explained the Scientometric is a discipline that allows the analysis of the characteristics of publications in each field of knowledge using different indicators. This study's objective was to examine international scientific publications about COVID-19 and dentistry. Methods: To find every document published in journals indexed in the Scopus database between December 2019 and February 2022, a specialized search approach was employed. SciVal (Elsevier) was used to export and analyze the study variables. Outcomes: A total of 2071 documents were obtained; the British Medical Journal (BMJ) was found to be the journal with the largest scientific output (107), and Oral Diseases has the highest impact among them with an average of 12.3 citations per publication. With 292 and 287 published papers respectively, India and the UK held the top two positions. The Brazilian university with the most effect and number of published papers (sixty-one) was Universidade de Sao Paulo. Conclusions: Scientific papers on COVID-19 and dentistry have been significantly rising during the pandemic. India and the UK are the countries with the largest scientific production, while international collaboration has the highest percentage of publications.

Okolie and Ogundeji, (2022) have highlight the Coronavirus disease has created an unexpected negative situation globally, impacting the agricultural sector, economy, human health, and food security. The research on COVID-19 in relation to food security and agricultural productivity was addressed in this study. In order to source research articles, Web of Science and Scopus databases were consulted, taking into

account the severe circumstances and scenarios brought about by the COVID-19 pandemic, including the lack of agricultural output activities and the harm they pose to food security systems. A total of 174 published publications were downloaded for additional research in BibTeX format. The authors utilized the search term "effects of COVID-19 on agricultural production and food security (ECAP-FS)" to find relevant papers using bibliometric novel methodologies for research published between 2016 and April 2021. The results showed an annual growth rate of almost 56.64%, suggesting that during the course of the study period, there was an increase in ECAP-FS research. However, there was variation in the research output on ECAP-FS; as of April 2021, 2020 accounted for 38.5%, while 2021 followed with 37.9%. By following our technique and using net2- VOSviewer, which is incorporated in the Rstudio software package, among other analyses, it was evident that one could perform cooperation network analysis. The proposed four-stage processes for combining two datasets for bibliometric analyses clearly demonstrated this. According to the study's findings, supply and demand shocks resulting from the pandemic's disruption of the agricultural food supply had an adverse effect on each of the four pillars of food security.

Qi et al., (2022) have described in recent decades, the positive youth development approach has gained momentum among researchers and practitioners who focus on youth potential and adolescent wellbeing. This study uses CiteSpace to gather, assess, and analyze 1435 scholarly papers on positive youth development (PYD) from the Web of Science. The number, distribution of subjects, nation, and institution of published papers are first shown, and network analysis is then used to examine the author collaboration network involved in PYD. The findings indicate that multidisciplinary collaboration is progressively increasing and that Western countries' authors and institutions are the primary sources of PYD research. Researchers looked into research hotspots in the field using document co-citation analysis. Moreover, burst detection was used to identify research gaps as well as the frontiers and hot themes in PYD from 2010 to 2019. The results offer avenues for future PYD research in the area of practical quality of life studies.

Santos et al., (2022) the concept of mapping and analysis of scientifc knowledge makes it possible to identify the dynamics and/or growth of a particular field of

research or to support strategic decisions related to different research entities, based on bibliometric and/or scientometric indicators. But as scientific production grows exponentially, it becomes more and more important to analyze this massive body of work in a methodical and data-driven manner. In order to extract implicit knowledge from scientific production bases, a data-oriented methodology combining data analysis, machine learning, and complex network analysis techniques, together with a data version control (DVC) tool, was proposed in this work. Furthermore, a case study in a COVID-19 papers dataset—which included 199,895 articles published on arXiv, bioRxiv, medRxiv, PubMed, and Scopus databases—validated the technique. The findings, which highlight the most active nations and the most frequently studied issues during each pandemic phase, point to the viability of the suggested methodology. Thus, this research has the capacity to help and broaden the scientific community's strategic decision-making in order to gather information that aids in the fight against the COVID-19 pandemic.

Atlasi et al., (2021) in this study totally, 56,402 records and 309 hot papers were retrieved. 3 of these hot papers were in endocrinology category. Original papers were the most popular kind of publication, with editorial papers coming in second. China was the second-most-published country after the United States. According to the ranking, the British Medical Journal and the Journal of Medical Virology were the top two sources. "The Lancet" was the top-ranked journal that published highly-cited papers, while "Harvard University" was the top organization with a high proportion of scientific publications. The body of knowledge about COVID-19 is expanding quickly and at a rapid rate. In this regard, it is necessary to periodically review these publications and publish the findings in order to utilize the data for more efficient management of upcoming research projects, emphasizing the gaps in the literature and producing more citation-worthy materials. Additionally, funding should be allocated for more critically needed research rather than duplicating work. This would be beneficial for the prevention, management, and treatment of COVID-19, which is currently one of the most talked-about issues worldwide.

Aviv-Reuven and Rosenfeld (2021) have examined in recent months the COVID-19 (also known as SARS-CoV-2 and Coronavirus) pandemic has spread throughout the world. In parallel, extensive scholarly research regarding various aspects of the

pandemic has been published. In this study, we examine how the epidemic has affected the patterns of biomedical publishing. We examine changes in the number of publications published in preprint servers and peer-reviewed journals, the average time it takes for papers submitted to biomedical journals to be accepted, the diversity and volume of international (co-)authorship in these papers, and any potential relationships between these changes and journal metrics. Two methods are used to investigate these potential changes: a short-term analysis that looks at changes in the first six months of the outbreak for both COVID-19-related and non-COVID-19-related papers, and a longitudinal approach that looks at changes in relation to the preceding four years. Our findings demonstrate that the pandemic has had a significant impact on all analyzed accounts of scientific publications thus far: Significant increases in publication volume have been observed, and nearly all of these can be directly linked to the pandemic; COVID-19 papers have a noticeably shorter mean time to acceptance than non-COVID-19 papers, which has (partially) resulted in a reduction in international collaboration for COVID-19 papers. These modifications may result in a reduction in research in biomedical fields unrelated to COVID-19 and a decrease in the rate of international cooperation as the pandemic spreads.

Bachalapur MM et al., (2021) have explained. The purpose of this paper is to assess the publications and citations that the professional institutions of BLDEA, which are based in Vijayapura, Karnataka State, India, under the auspices of the BLDE Associations, have received over the course of ten years, from 2011 to 2020, as shown in the Scopus Citation Database. In accordance with the goals of the current study, a total of 910 research articles and citations with all metadata have been downloaded and examined. Four thousand two hundred forty-seven citations were sent to these three professional institutes. The goal of the study is to present all metrics for examining the research output and citations of the professional institutions of BLDEA in the fields of engineering and technology, pharmaceutical science, and health sciences.

Colavizza et al., (2021) pointed in the COVID-19 pandemic unfolds, researchers from all disciplines are coming together and contributing their expertise. A dataset of COVID-19 and coronavirus papers called CORD-19 has been made public along with

requests for assistance in mining the data it includes and the development of tools to facilitate more efficient searches. From a scientometric standpoint, we examine how the papers that are part of CORD-19 are defined. Comparing CORD-19 to the Web of Science database, we see that it offers nearly comprehensive coverage of studies on coronaviruses and COVID-19. In addition to studies specifically focused on COVID-19 and coronaviruses, CORD-19 also includes study on viruses in general. A select number of clearly defined research areas are the main focus of CORD-19 publications: coronaviruses (mainly SARS-CoV, MERS-CoV, and SARS-CoV-2); public health and viral epidemics; virus molecular biology; influenza and other viral families; immunology and antivirals; and clinical medicine. Publications from CORD-19, particularly letters and editorials, are disproportionately well-liked on social media in 2020. Although we wholeheartedly support the CORD-19 program, it's crucial to understand that its scope goes beyond studying coronaviruses and COVID-19.

Cunningham et al., (2021) have purposed to make novel coronavirus SARS-CoV-2 and the COVID-19 illness it causes have inspired unprecedented levels of multidisciplinary research in an effort to address a generational public health challenge. We do a scientometric analysis of COVID-19 research in this article, focusing on the type of interdisciplinary collaboration that this pandemic has encouraged. It has been demonstrated that more multidisciplinary collaboration results in greater scientific effect at the expense of higher coordination costs. Therefore, to evaluate the scope and diversity of collaboration in COVID-19 research, we take into account a corpus of more than 166,000 COVID-19-related papers, which we compare to non-COVID-19 controls both before and during the pandemic. We demonstrate that research teams affected by COVID-19 are not only considerably smaller than those affected by other viruses, but also more heterogeneous. Moreover, we discover that COVID-19 research has led to a rise in the multidisciplinary composition of authors in the majority of scientific fields, suggesting that COVID-19 has assisted in reducing some of the barriers that typically separate different disciplines. In conclusion, we identify several intriguing fields of multidisciplinary study during COVID-19 and suggest techniques for illustrating the nature of multidisciplinary cooperation that might be useful outside of this pandemic.

Hussain et al., (2021) the study's main purpose was to map Cerebral Palsy (CP), a bibliometric analysis of the highly cited research output, 1990-2020. All of the CP-related records were retrieved via the web of science database. The reviews' Impact Factor was obtained from the IF2019, the 2019 edition of Clarivate Analytics' Journal Citation Report (JCR). Based on a database of the most active writers and organizations, the Hirsch Index (h-Index) was created. The co-citation networks among productive writers of highly cited publications were utilized by ScientoPy, VOS viewer, and Biblioshiny. The primary conclusions of the study showed that Australia's Royal Children's Hospital Melbourne was the most productive institution and that the United States was the nation that contributed the most publications. The greatest number of articles on CP research have been published in the fields of developmental medicine and child neurology. It might make it possible for later studies to look into other areas, such mapping the output of CP literary study, which is a bibliometric analysis of the publications of papers in a specific nation and continent.

Malik et al., (2021) have focused on Infectious diseases remain a complex, recurring, and challenging public health hazard. Coronaviruses have had a variety of negative effects on socioeconomic situations, mobility, and health. Although the impact of pandemics and epidemics has been significant, there is a dearth of literature on the comprehensive study performance of coronaviruses over time. The objective of this work was to conduct a scientometric analysis of literature on coronaviruses, particularly COVID-19. Information about research on coronaviruses was taken from the Web of Science (WoS). 288,846 publications of all kinds were included and retrieved. A thorough analysis examining a variety of variables was conducted using the "R-Bibliometrix" tool in order to quantify the number and caliber of the articles. Over time, a general pattern of increase was noted, with the United States and China leading the way, followed by Europe, the United Kingdom, and a small number of other developed nations. Only the first six months of 2020 added another around 46.5% of the total documents, with the remaining two decades contributing approximately 39.5% of the total. In the previous two decades, shorter spikes of higher post-pandemic publications followed by declines in output were noted; these indicated a lack of continuity and were described as "a research epidemic following a disease epidemic." Articles accounted for 53.4% of all publications. Leading sources included the Journal of Virology, British Medical Journal (BMJ), and Virology; more

recently, the Lancet and BMJ have contributed more. With a few notable outliers, mostly from impacted areas, comparable patterns of top authors were seen in terms of output, impact, partnerships, funding sources, and affiliations overall. The majority of documents (>89%) from the top 20 countries indicate a deficiency in global efforts. It was shown that affluent countries were the key participants in networking, with resource-constrained nations contributing less and possibly needing more collaboration. The spike in publications since the COVID-19 pandemic is unprecedented, steep, and expanding quickly. The latest COVID-19 pandemic is highly recommended by the authors as a call for ongoing, more collaborative, and group-based international research.

Mehmet Oguzhan Ay et al., (2021) have presented the Coronavirus disease 2019 (COVID-19) caused by the novel coronavirus SARS-CoV-2 that was declared as a pandemic has been the main subject of research all over the world. Research on COVID-19 vaccinations in particular has given everyone hope. In order to determine the present state of the COVID-19 vaccination literature, we set out to analyze the full body of material using the Web of Science© Core Collection Database. On January 20, 2021, we entered the terms "COVID-19" and "vaccine" into the Web of Science© Core Collection Database. Investigations were conducted into Web of Science categories, document kinds, organizations, funding agencies, authors, journals, nations, languages, and fields of study. In all, 2,765 papers totaling 24,202 citations were included in the analysis. The majority of the publications were first-authored works. The top three categories were experimental medicine research, general internal medicine, and immunology. Harvard University, University of California System, and University of London were the three most prolific universities. The most publications were by Dhama K., then by Mahase E. and Baric RS. With the most publications, the Journal of Biomolecular Structure Dynamics held the top spot. The publications were written in English for the most part. China and India were the next most productive nations, after the United States of America. The field of vaccine research is expanding and is vital to the effort to combat COVID-19. The future viewpoint may be discovered by researchers with the aid of in-depth examinations of vaccination articles.

Pal (2021) has examined the scholarly output of the new coronavirus research has

been proliferating. In just five months, 14,588 scientific papers about COVID-19 and nCoV-2 have been produced (as indexed in Scopus on May 31, 2020). There is a great deal of interest in comprehending the research landscape of this recently organized field as a result of this knowledge explosion. In this work, scientometric aspects of the novel coronavirus (2019-nCov) research are illustrated through quantitative features of the published dataset. The results show how important the pace of publication increase (1600%) is to the researchers' collaborative effort to combat the longest RNA viral sequence. In fact, they are already responding with up to 100 pieces a day on average. Numerous academic publishers have opened up their preprint servers to the public so that the articles are instantly accessible, even with Open Access enabled. The scientific articles have been published by 240 academic publishers in more than 500 periodicals. About 25% of the studies were funded by 300 funding bodies, despite the fact that the top ten publishers accounted for about 70% of the articles. Prominent journals include Lancet, Nature, BMJ, JAMA, JMV, and NEJM. The bulk of contributions have been made to the field of medical science, with an emphasis on critical care, emergency medicine, virology, immunology, epidemiology, pharmacology, and public health. On the other hand, terminology like "viral pathogenesis," "pandemic," "asymptomatic," "quarantine," "pneumonia," "human," "disease severity," "clinical trials," and "risk" and "mortality" are closely related. According to the report, the USA, China, Italy, and the UK are home to the majority of academic hubs. Huazhong University (China), Tongji Medical College (China), Harvard Medical School (USA), University of Milan (Italy), INSERM (France), and UCL (UK) are among the highly prolific universities. Collectively, the G7 nations generated half of the world's nCov-2 research output. Additionally, a promising trend of collaborative research across several nations and fields was identified, with extremely significant values for CI (6.46), DC (0.79), and CC (0.59). It looks at the writers' geographical diversity and uses co-authorship occurrences to display their connections. Lastly, it examined the papers' effects to highlight the most significant contributions made by the most recent coronavirus research.

Rehman et al., (2021) has analyzed The Journal of Family and Community Medicine (JFCM) is a peer-reviewed open access journal published by the Saudi Society of Family and Community Medicine. The purpose of this review is to use scientometric technique to examine the scientometric attributes of articles published throughout 27

years, from 1994 to 2020. We obtained bibliographic records of publications published between 1994 and 2020 from the Medline PubMed and Web of Science databases. Software such as Biblioshiny, CiteSpace, and VOSviewer were used to evaluate the data. There were 1442 authors contributing a total of 648 manuscripts, or an average of 2.22 authors per manuscript and 24 manuscripts annually. 2,693 citations were added to all the manuscripts, with an average of 4.15 citations per manuscript. The single-author pattern accounted for 48% of the submissions; multiauthor publications had a greater ratio of citations. All 20 of the top contributing writers were Saudi nationals. The evaluation emphasized the nations and institutions that contributed the most. Additionally, there was a bibliographic coupling of nations, organizations, co-occurrence of keywords, and co-citation of journals. For the past 27 years, family medicine researchers have been able to share their academic and scientific communication through the valuable platform offered by the Saudi Arabian journal JFCM. Both the number and frequency of journal articles have increased over time. Despite submissions from all across the world, the majority of the journal's contributions came from Saudi Arabia.

Şenel and Topal (2021) stated that our main source for this study was Web of Science Collection database. Every item released between 1980 and 2019 was incorporated. A global production distribution map was created using scientometric networks and coronavirus literature. The most productive nations were the United States, China, Germany, the United Kingdom, and the Netherlands. Almost all nations, with the exception of a few in Asia and Africa, have published works in the coronavirus literature. While the United States and developed European nations were the main source countries in the 1980s, the virus was only recognized as an animal disease in the literature and its biological and genetic structure was studied; however, because the SARS outbreak started in southern China, China began to contribute significantly to the literature on coronaviruses in the 2000s. The majority of the top cited papers during this time frame have something to do with SARS and the ACE2 protein.

Sofik and Rahman (2021) focused the Information retrieval (IR) is the procedure of obtaining information system resources relevant to an information need from collecting those resources. Over the years, information retrieval (IR) has grown

significantly and taken the lead in terms of information access methods. This study's major goal is to investigate international scientific papers on information retrieval (IR), map out current worldwide trends, and visualize this cutting-edge field of study. By doing so, it will be possible to suggest future research directions based on dynamic observations. The study examines the Web of Science core collection database, which covers published items from 2006 to 2020, using a bibliometric analytical approach. Using ScientoPY, Biblioshiny, Histcite, and VOS-viewer, the paper also created a graphical visualization and knowledge mapping of the bibliographic literature by mapping journals, keywords, and institutions with bibliographic co-occurrence of terms, subject, and co-citation analysis. The study thoroughly assesses the state of IR today, which will inspire the researcher to create creative projects to support efficient and ongoing research.

Surulinathi et al., (2021) have studded Based on a number of factors, including the publications share and rank, institutional profiles of chosen institutions, international collaboration profile and major collaborative partners, communication patterns in national and international journals, highly cited papers, and most productive authors, the study assesses India's performance in relation to its publication output in COVID-19 during 2020–2021. The study makes use of two years' worth of COVID-19 publishing data that was obtained from the international multidisciplinary bibliographical database Web of Science. It's interesting to note that the most prolific writers are affiliated with various institutions: Banaras Hindu University (171 Citations), Indian Institute of Technology (54 (349 Citations), Dr. DY Patil University (382 Citations), Post Grad Inst Med Educational & Res (171 (666 Citations), and All India Institute of Medical Science (489 Publications, 1266 Citations). With 115 publications and 372 citations (H-Index 10), Kumar A from Armed Forces Med College, Dept. of Ophthalmology, Pune (Maharashtra state) is the most productive author. Wiwanitkit from Dr. DY Patil Univ, Pune (Maharashtra state) has 91 publications (411 citations) (H-Index 11), Kumar S from Jai Narayan Vyas Univ, Jodhpur (Rajasthan state) has 85 (421 citations) (H-Index 10), Gupta S has 57 (209 Citations) (H-Index 7), Singh has 53 (175 Citations), and Sharma S has 52 (359 Citations).

Ahmad et al., (2020) the paper examines and evaluates the global literature trend on

"Coronavirus Disease" in terms of research publication output as indexed in Web of Science's Science Citation Index Expanded (SCI-E) between 2011 and 2020. According to the study, till March 20, 2020, 6071 research reports on coronavirus disease had been published. The different scientometric elements of the research papers released throughout the investigation period were examined. The study highlights the different elements of the literature on coronavirus diseases, including the distribution of the literature by year, the relative growth rate, the literature's doubling time, the literature's geographical distribution, organization, language, form, most prolific writers, and source. 2019 saw the most number of publications released, while 2020 saw the fewest research articles recorded. Furthermore, while doubling time lowers, the relative growth rate increases steadily. The majority of research publications are published in the English language and take the form of research articles. The United States of America has contributed the most to the literature on coronavirus diseases.

Asnani and Vyas (2020) have described. An increasing trend and thrust topic in scientometric research is the quantitative study of publication aspects and citation analysis, which generates both quantitative and qualitative research for characterizing the output of cited works. This study presents a qualitative and quantitative analysis of the countries, institutions, and journal contributors. This study examined 1872 papers totaling 15711 citations in order to determine the productivity of the journal "Scientometric" between 2015 and 2019. The growth rate of publications and the influence of citations, author productivity, authorship patterns, and the most pertinent distribution of contributors by country and institution are all covered in this study. For data analysis, information retrieved from the Web of Science was utilized, along with the R program Bibiblimetrix, and VOSviewer for cooperation networks. To connect the relationships between the nation, authors, and keywords, a Sankey diagram was employed.

Dadhe et al., (2020) To understand This scientometric study, which examined the global research trend of coronavirus scientific publications from 2002 to 2019, was based on scientometric indicators, including the growth of publications, productive authors and institutions, collaboration between authors and institutions, research hotspots, and citation patterns. The dataset for the study consisted of 8541 records that

were downloaded from PubMed and were analyzed with the use of Bibexcel and VOSviewer. The results show that while coronavirus research was evident, it was not significant. There was also a discernible relationship between the quantity of articles and the outbreaks of MERS in 2012 and SARS in 2002. Throughout the study period, it was discovered that the publication tendency was not constant. The results show that there was little institutional collaboration. Molecular sequence data, animals, people, females, males, and mice are among the frequently used terms, according to keyword analysis.

Danell et al., (2020) The aim of this article is to examine how complementary and alternative medicine (CAM) has evolved as a field of study between 1966 and 2016. The publication pattern will be examined using bibliometric techniques, with an emphasis on journal articles and sources. In addition, we will map the overall content of the papers and examine how clinical research has evolved. The data set is made up of 105 216 publications from the Medline database with complementary therapies as their MeSH major theme. During the late 1990s, CAM research began to grow. Approximately 20% of the articles at the conclusion of the period were categorized as clinical trials, while not many were categorized by clinical phase. According to the Bradford law of dispersal, the bulk of core publications had a CAM focus. Based on SNIP values, these journals' influence was rather ordinary. Acupuncture therapy, mind-body therapies, and musculoskeletal manipulations were the most popular therapies, according to the MeSH term analysis. The largest single node was found to be acupuncture, according to the co-word analysis of the author's keywords. It was also feasible to identify a particular CAM cluster. CAM has grown into a narrow yet unique field of study. There are indications of both incorporation into other medical subfields and expertise in/about CAM. Moreover, there are indications of a nascent medical field, distinguished by limited and exploratory clinical investigations.

Danesh and GhaviDel (2020) have explained This practical research was conducted with an analytical methodology and scientometric tools. 5128 documents related to the coronavirus subject area that were indexed on the WoS between 1970 and 2019 make up the statistical population for this article. Excel 2016 was used to extract the keywords from MeSH and analyze them. According to data analysis, 2019 had the most citations and the highest level of research creation, respectively, in 2005 and

2019. The United States is the most producing nation, "Enjuanes L." is the most widely read author, and the University of Hong Kong is the leading coronavirus research organization over the past fifty years. The findings demonstrated a clear correlation between coronavirus outbreaks and the quantity of scientific publications published worldwide in this field. Scientific methods can be used to evaluate the caliber of the researchers' output in this field, and the h-index of the researchers has been impacted by their self-citations. The findings of scientific studies in strategic and important research areas, like coronavirus, must be known by health care researchers, policymakers, and planners in order to set more appropriate therapeutic goals, make better decisions, and deliver more efficient solutions in the shortest amount of time.

Haghani and Bliemer (2020) Every significant coronavirus outbreak in the twenty-first century have resulted in an abrupt and rapid increase in scholarly papers on the subject. Conversely, the 2019 outbreak of the Novel Coronavirus (COVID-19) has caused an unprecedented surge in scientific papers. The COVID-19 pandemic has raised attention to a worldwide crisis that has sparked tremendous scientific research. In less than five months, over 12,000 research items were indexed, and in less than seven months, over 30,000 things. By the end of 2020, the number might potentially exceed 80,000 if the current trend continues. It seems like COVID-19 research has become a focus point of attention across many academic disciplines, as the health crisis affects all facets of society. The present study examines and compares the scientometric features of the Middle East Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS), the two prior major coronavirus illnesses, are compared with the literature on COVID-19. Important terms appearing together, bibliographic coupling, and journal citations relationships, and international cooperation are the main points of interest. Recurring themes that were intriguing were found in all three literary works. Three unique cohorts of studies have been produced by all three outbreaks: (i) studies pertaining to public health response and epidemic control; (ii) studies examining the chemical makeup of the virus; and (iii) studies concerning treatment, vaccine, and clinical care. Studies in category (i) appear to have been among the first to appear, but overall, their number of citations has been lower than that of publications in the other two categories. Studies conducted using COVID-19 appear to have been published in a wider number of journals and subject areas. There are definite correlations found between the geographic sources of each

outbreak, the local geographic intensity of each outbreak, and the volume of research that emanated from those areas. Compared to SARS and MERS, COVID-19 studies also show the participation of writers from a wider range of nations. Examining whether the caliber and rigor of these articles have been impacted, given the rapidity with which the literature on COVID-19 is growing, is an intriguing aspect that merits more research.

Haghani et al., (2020) have analyzed the COVID-19 global pandemic has generated an abundance of research quickly following the outbreak. In just a few months, the scientific literature has already published over a thousand research on this subject. In this brief review, we examine these research' macro-bibliometric features as well as those that deal with coronaviruses in general. Additionally, by doing a scoping review of the COVID-19 literature, we pinpoint the primary safety-related aspects that these investigations have focused on up to this point. Our findings demonstrate that, in a variety of research domains, and aside from the medical and clinical aspects such as the safety of vaccines and treatments, the scientific community has focused most of its attention on COVID-19-related issues related to patient transport safety, occupational safety of healthcare professionals, biosafety of laboratories and facilities, social safety, food safety, and, in particular, mental/psychological health and domestic safety. Our investigation also reveals a number of potentially serious safety issues brought on by this worldwide health catastrophe, which may require additional attention but have received little scientific attention thus far. These cover topics including supply-chain security, economic safety, and cyber safety. The aforementioned results underscore the need for an all-encompassing, multidisciplinary strategy and collaborative scientific endeavour to comprehend and alleviate the diverse safety consequences of this emergency, which surpass the biomedical hazards. Furthermore, having a comprehensive safety-scientific understanding of the COVID-19 situation will help us prepare better in case of another pandemic.

Kagan et al., (2020) analyzed on COVID-19 is the most rapidly expanding coronavirus outbreak in the past 2 decades. Results: Here, we examine the amount of research done on previous coronavirus outbreaks, specifically SARS and MERS, in comparison to other infectious diseases by analyzing >35 million articles from the past 20 years. Our findings highlight the understudied nature of previous coronavirus

outbreaks when compared to other viruses, as well as the fact that the volume of research on emerging infectious diseases is very high during an outbreak and sharply declines after the disease is contained. This can result in insufficient research and limited investment in order to fully comprehend novel coronavirus management and prevention. Conclusions: Regardless of how the current COVID-19 outbreak plays out, this research will be done regardless of how well-funded

Kappi et al., (2020) have described On February 26, 2020, the Web of Science database was searched for papers about the corona virus (COVID-19) that were released between 1997 and 2020. Since the metrics are constantly changing, it was done on the same day to minimize any potential bias resulting from database updates. Publications that were errata were not included, but all other publishing categories were taken into account. Publication year, publication kind, international collaboration patterns, research institutes, journals, impact factor, h-index, language, and times cited are some of the criteria used in the analysis. Globally, 12612 scientific publications on the COVID-19 virus have been published. The articles related to the corona virus (COVID-19) came from 25 different nations and territories, demonstrating the global dissemination of corona virus (COVID-19) research. With 4524 papers produced over 32 years, the USA was the largest contributor, followed by Peoples R China with 2667 pieces. With an average of 9.01 citations per publication, the total number of citations for these papers has now reached 8,015. Publications related to the corona virus (COVID-19) have an h-index of 48. The highest h-index is likewise 32 for the USA, followed by 26 for KSA and 22 for the UK. Out of all the articles published for each country, the Netherlands produced the highest percentage (72.7%) of publications including international research collaboration, followed by the UK (71%), Germany (691%) and other countries.

Laksham et al., (2020) this study presents the overview of coronavirus research output from 1989 to March 2020 from a worldwide level was provided by these analyses, which also included information on the most prolific journals, year-by-year growth in research, and global publishing shares and patterns. 7381 papers were found when data was pulled from the Web of Science citation database with the search terms "Coronavirus" OR "COVID 19" and restricted to Open Access papers between 1989 and 2020. With 561 publications published in 2019 and 848 citations, it was the year

with the most publications. Thus, it can be said that the output of joint authors outweighs that of (SAP) Single Authored Publications in this paper. Based on a global literature study, 127 nations were represented among the 7381 publications. The most productive nation is the United States of America (USA), with 2801 publications (37.9%) and 107738 citations. In the future, India (80) needs to make progress in the field of coronavirus research. The research publications that are published in open access, peer-reviewed journals will have a worldwide impact on scientists, countries, and academic institutions. These contributions will support coronavirus researchers and assist the scientific community in obtaining information needed for the field.

Mukherjee (2020) has described. The COVID-19 (n-CoV) virus is a member of the huge viral family known as "Coronaviruses," which can infect humans and animals with respiratory and gastrointestinal diseases. The goal of this study is to comprehend the patterns in worldwide research during the past seven decades in disorders connected to coronaviruses. As a result, it aims to offer a knowledgeable evaluation of the relevant research. The "acute respiratory syndrome" and the "gastroenteritis virus" and "mouse hepatitis virus" (mhv) fusion and penetration processes of this virus were the main topics of research on this illness. The two most productive nations were the USA and China. While collaborative research in China tended to be global, it was primarily conducted on an intra-national level in the USA. The study on coronavirus-related disorders that has been conducted in these two most populous nations is expounded upon and highlighted in this publication. Citation analysis reveals some fascinating patterns. Even though an article earned an average of 27.76 citations, 10% of citations came from only the top 56 (0.34%) articles, indicating that only a small percentage of all publications received international attention. There may not be as much collaborative research team effort in this topic as the weak collaboration link between highly cited writers suggests. This field of study has been studied since the early 1950s. It is not unexpected that, compared to earlier times when the illnesses brought on by this virus were more isolated, there has been a recent surge in the amount of study being done in this field. It offers hope that this carefully focused research being conducted in various nations will open up new avenues for comprehending coronavirus-generated illnesses, such as the current n-CoV, which is a necessary prerequisite for creating vaccines to prevent coronavirus-associated diseases and control them.

Odone et al., (2020) Context and goal of the work: To consider the trends, substance, and caliber of COVID-19-related scientific publications. Specifically, to report on the methodical screening, quantitative evaluation, and critical appraisal of the first 10,000 scientific publications on COVID-19 that were published, and to evaluate the ways in which research priorities and public health requirements were met by scientific outputs. Methods: A thorough research plan was created in order to regularly and methodically retrieve every study that has been published on COVID-19. We retrieved the following from the included studies: study design, country of implementation, and bibliometric factors. The publications were categorized into 25 predetermined COVID-19-related categories, and the scientific outputs were compared to the academic publishing rankings of the countries and the COVID-19 load. Findings: During the period of January 20, 2020, to May 7, 2020, 10,000 scientific publications were published on COVID-19, making up 2.3% of all scientific production during that time. 33% of the participants concentrated on COVID-19 clinical care, with minimal attention paid to the designated research priorities. More than 60% of the papers contained opinion pieces rather than real research. Papers were published in 1881 journals, however only 8% of journals contained half of the scientific output. Of all scientific production, the United States contributed one-fourth, followed by China (22.2%) and Italy (9%). Conclusions: This is the largest amount of research ever published by an academic journal on a single topic, and it is likely to bring about significant changes in the production and dissemination of science, possibly detracting from its primary goal of guiding clinical and public health practice and decision-making.

Patil (2020) The global COVID-19 pandemic has presented numerous concerns to the international community. Numerous nations and regions were placed under lockdown. The scientific communities have increased their research and development (R&D) efforts in an effort to contain the ongoing pandemic. A large number of publications have removed their paywalls and made the material on coronaviruses freely accessible. The current study examined 2184 COVID-19 papers that were indexed by the Dimensions database between January 1, 2020, and March 27, 2020. The research discovered a pattern of multi-authorship in COVID-19 material. China is a highly referenced country, according to the data. It was discovered that Hiroshi Nishiura has a robust network of collaborators. It has been determined that "Nature" is the most

productive journal in terms of citations. It's interesting to note that the researchers favored sharing and distributing their COVID-19-related preprints via "medRxiv," the Preprint Server for Health Sciences. It is discovered that China is home to the vast majority of the active institutions that produce papers connected to COVID-19. It was discovered that the universities of Oxford, Hong Kong, and Fudan have a robust network of collaboration. The Lancet and the New England Journal of Medicine are the two journals that lead the list of highly cited references. According to the analysis, the keywords coronavirus, pandemic, data, ncov, etc. appeared frequently in the COVID-19 papers.

Radha (2020a) Scientometric study is one of the most efficient methods of quantitative evaluation of the scientific publication. One type of frequent virus that occasionally affects both humans and animals is the coronavirus. There are now seven different types of corona, however a newer coronavirus known as novel corona (COVID-19) is spreading quickly among people. This study aims to learn about the latest coronavirus research publications from the scientific community, particularly after December 2019, when the true spreads began and the lockdown had an influence on the global economy. The web of science provided the data for this study for the months of March in the years 2018, 2019, and 2020. This study aims to determine the publication pattern, growth of publications year over year, research output by country, and most cited papers. Additionally, the research output from China and Wuhan University in the publications, as well as Wuhan University's involvement in this research publishing during the era, are the main topics of this work.

Radha (2020b) This study looked at the professors at Thiagarajar College of Engineering (TCE), in Madurai, Tamilnadu, India, and their research publications. The Scopus and Web of Science databases were used to gather the data for this study from 2014 to 2018. There were 437 publications published and indexed in the Web of Science and 1270 papers published and indexed in Scopus throughout a five-year period. The h-index, i-10 index, average citation per year, cited rate, percentage of cited and uncited papers, true average citation, etc. may all be calculated using an Excel sheet, as demonstrated in this work. In addition to the previously mentioned study, this paper identifies Scopus and Web of Science papers that overlap.

Ram (2020) Coronavirus (CoV) is large family of virus and is endemic in human and animal causing respiratory and intestinal infection. Due to its recent outbreak at the beginning of 2020, it has become a challenge in the China region. The Wuhan area of China is experiencing a deadly COVID outbreak that is confining its citizens to their homes. When Middle East Respiratory Syndrome (MERS-CoV) spread throughout the Middle East and severe acute respiratory syndrome (SARS-CoV) spread throughout Hong Kong in 2003, both types of CoV became epidemics. This bibliometric analysis uses the SCOPUS database to try and track the patterns in research on "Coronavirus" over a half-century period. Using the term "coronavirus," the study was conducted and its annual growth, productive countries, institutes, authors, journals, highly cited articles, and research focus were all examined. It is discovered that there were two peaks at the times when a lot of articles came out. The majority of research articles (31.67%) came from the USA, with University of Hong Kong emerging as the most productive institution. The most articles about CoV have been published in the Journal of Virology.

Ram and Nisha (2020) in this paper, based on information obtained from the SCOPUS interdisciplinary database on "Coronavirus" research published between 1970 and 2019 and 2020, an analysis of 806 highly cited publications was conducted. The fields title, abstract, and keyword were searched for and data was retrieved using the term "Coronavirus." In addition, a review of the highly cited publications' citation life cycle, author performance, institutional performance, national performance, and journal performance was done. A total of 18,116 articles were published over the time through 2019; 1221 items were published in 2020 through March 30, 2020. In coronavirus research, publications with over 100 citations were considered highly referenced. 806 articles with more than 100 citations were taken into consideration for the data analysis based on this criterion. The majority of the highly cited papers came from American publishers. The majority of the highly cited papers were published in the "Journal of Virology," which was the most productive journal. These widely cited works were released from 1973 to 2016. The most productive institution was the University of Hong Kong in Hong Kong, which also published the most highly cited articles. The most prolific author, having authored the most highly cited pieces, was Kwok Yung Yuen of the University of Hong Kong in Shenzhen. Citation trends for the seven articles revealed traits of "sleeping beauty."

Sahoo and Pandey (2020) Using scientometric indicators such as author analysis and network, h-index and citation, citation growth and relative citation impact (RCI) of prolific countries and influential institutions, degree of collaboration (DC), collaboration coefficient, and modified collaboration index (MCI) in the field of coronavirus and COVID-19 research, this study aims to assess the growth of scientific literature in this area. Design, procedure, and strategy The authors used methods to conduct a systematic search utilizing keywords associated with the subject domain under study in order to retrieve the literature data from the Scopus database between 2000 and 2020. 15,297 records in total were taken into consideration for the literature analysis, taking into account the actual noteworthy expansion of this field of study. The scientometric analysis of these papers was reported in this study. Moreover, the collaboration pattern has been explained using statistical relationships. Using the visualization application VOSviewer, the co-author network is built. Results: This analysis discovered that 8,195, or 53.57%, of the research papers were published on the open access platform. The researcher discovered that the Journal of Virology was their top choice, providing about 839 (5.48%) articles. The United States and China account for the majority of research output, with the University of Hong Kong producing the most research papers (547, or 3.58%). Medicine accounts for 49.70% of the research documents published, with immunology and microbiology coming in second and third, respectively, and biochemistry, genetics, and molecular biology coming in third and second, respectively. Since January 2020, there has been an unprecedented global distribution of research money together with an unprecedented proliferation of papers on COVID-19. Limitations and consequences of the research: This study only looks at 15,297 research outputs that have been indexed in the Scopus database between 2000 and 2020 (up until April 1, 2020). As a result, research is not limited to documents published in any other channels or sources that are not covered by Scopus. Practical implications: It will help researchers and practitioners throughout the world comprehend how the scientific literature on COVID-19 and the coronavirus is growing and find possible collaborators. Originality/value: This study is important in the current context for recognizing the increase of scientific literature in this subject and the evolution of this topic of research worldwide, given the global effect and social suffering caused by the COVID-19 pandemic. The research findings are helpful in identifying important research trends from publications and advancements in the COVID-19 and coronavirus fields.

Singh et al., (2020) COVID-19 is a persistent menace to humanity that has shook the entire world and for which there is currently no cure. There is a new coronavirus called COVID-19. Research into COVID-19 is required in order to develop a remedy or vaccination. As of 4 April 2020, 208 nations have reported 1,056,159 confirmed cases and 57,206 deaths. Therefore, bibliometric analysis of COVID-19 literature taken from the Web of Science database for the period of January 1, 2020, to March 31, 2020, has been done in order to examine the research trend on COVID-19. Based on the investigation, 489 documents were found, with BMJ and Lancent publishing the most articles on COVID. Mahase E. and Drosten C. are the two leading authors in this topic. Huang Cl, 2020 DOI:10.1016/S0140-6736(20)30183-5 is the most cited paper. China has contributed the most, with University of Hong Kong in Hong Kong ranking first among all organizations.

Sudhier and Dileepkumar (2020) looks at 25,132 Indian experts' contributions to biochemistry research that were published in the Web of Science over a ten-year period (2004-2013). It was discovered that the average yearly growth rate for biochemistry research is 36.84 percent, and it is increasing steadily. In Indian biochemistry research, team research is more common than solo research, with several authors contributing to 96.46% of publications. Throughout the study period, it was noted that the co-authorship index value was generally rising, ranging from 93 to 105. Contributions from journal articles account for 894.3% of the total, with reviews coming in second with 7.14%. The majority of Indian research collaborations (2.49%) are with US researchers. According to the geographical distribution, Delhi, Uttar Pradesh, and Tamil Nadu are at the top of the list. The study reveals that the top two authors in the field are L. Pai (37), and C. Abdul Jaleel (58). The top-ranked publication, "Plos One," released 296 papers throughout the study period. The majority of publications (50.26%) are contributed by academic institutions, with research institutes coming in second (28.24%). The observed author productivity distribution of Indian biochemistry research was not found to be consistent with Lotka's law.

Surulinathi et al., (2020a) have analyzed 433 publications in all were published on COVID-19, according to the Web of Science database, and they garnered 52567 citations between 1989 and 2020. There were 121.4 citations on average for each

publication. The research reached its zenith in 2020 with 97 publications (of which at least 500 gained citations), and the year 2020 also had the greatest number of citations (14623). It was noted that publications were growing exponentially. This might be related to a recently established field of study that aims to explain the COVID-19 vaccination phenomenon. The scientists published 2906 multi-author collaborative works and 18 single-author works. There is a 7.05 Collaboration Index (CI) and 15055 Citing Articles (References).

Surulinathi et al., (2020b) this article presents a Scientometric analysis of COVID-19 Vaccine research Literature indexed by Web of Science. Future protective and adaptive policy planning requires a thorough evaluation of research effort. The current study's goal was to evaluate the literature on COVID-19 vaccine research. Using documents on "COVID-19 literature" from Scopus, a scientometric approach was used. The years of study were 1971–2020. 7181 papers in the literature on coronavirus vaccines were discovered using the search criteria. In 2020, the number of publications increased to 4402. The most articles (203) came from documents published in the Vaccine magazine, followed by the magazine of Virology (104 publications) and Nature (96 publications). The United States, with 2178 (H-Index-114), China, with 1068 (H-Index-75), India, with 678 (H-Index-26), and the United Kingdom, with 614 (H-Index-53) are the most productive nations.

Thelwall (2020) Prior studies with other coronaviruses contributed to the speed at which biomedical professionals were able to identify and characterize COVID-19. Predicting the anticipated spread and mortality of COVID-19 was also made easier by early epidemiological comparisons with Middle East Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS). Using Mendeley reader counts as early academic impact data, this paper evaluates whether academic interest in earlier coronavirus research has translated into interest in the main source material. The findings validate that there were unusually large spikes in Mendeley readers in April and May 2020 related to SARS and MERS research conducted between 2008 and 2017. However, compared to original investigations of SARS or MERS, studies that use these diseases as benchmarks for COVID-19 or that learn COVID-19 lessons from them have attracted far more scholarly attention. Therefore, when new diseases are discovered, research that interprets previous relevant research seems to be

especially useful to help researchers grasp its significance in the current setting.

Varshney et al., (2020) 1106 answers in all, coming from about 64 different cities across the nation. At least one item was missing from 453 of these replies, which were not included in the analysis. With a male to female ratio of 3:1 and a mean age of approximately 41 years, approximately 22% of the respondents were employed in the health care industry. Significant psychological impact was reported by about one-third of respondents overall (IES-R score > 24). Female gender, concomitant physical illness, and younger age were all associated with higher psychological effect. Although they did not achieve statistical significance, the presence of physical symptoms and contact history indicated a stronger psychological impact.

Zyoud and Al-Jabi (2020) According to the bibliometric research, 19,044 papers about COVID-19 were published on Scopus between December 2019 and June 19, 2020, the early stages of the pandemic. There were 9140 (48.0%) articles, 4192 (22.0%) letters, 1797 (9.4%) reviews, 1754 (9.2%) editorials, 1728 (9.1%) notes, and 433 (2.3%) other types of publications out of all of these. The United States of America (4479; 23.4%) was the country with the most publications on COVID-19, followed by China (3310; 17.4%), Italy (2314; 12.2%), and the United Kingdom (1981; 10.4%). The most productive journal was British Medical Journal. The universities that published the most COVID-19 research were Harvard Medical School, Tongji Medical, and Huazhong University of Science and Technology. "Clinical features studies," "pathological findings and therapeutic design," "care facilities preparation and infection control," and "maternal, perinatal, and neonatal outcomes" are among the most often researched subjects in COVID-19. The fast-developing fields of COVID-19 research, where significant research has already started in earnest during the early stages of the outbreak, may be reflected in this bibliometric analysis. The results presented here provide new insight into the significant advancements that will be made in the near future in the hot themes of COVID-19 research, such as clinical feature investigations, pathological discoveries and therapy design, infection control and facility preparation, and outcomes for mothers, perinatal, and newborns.

Garg and Kumar (2019) Approximately 45% of the 595 documents created by Dr. Hari Chand Sharma, according to a study done on them, were published as journal

articles. The majority of the research articles were published in medium impact factor journals that were founded in the UK, the USA, and the Netherlands. The majority of his works have been coauthored by him with other scientists, and his high collaboration coefficient (0.66) and collaboration rate of 1.00 over a significant number of years imply that he has collaborated with over 1,000 writers. His most productive years were between the ages of 58 and 60. The large number of citations for each work suggested that Dr. Sharma's research outputs were included in mainstream science.

Gupta and Dhawan (2019) focused on both quantitative and qualitative aspects, a summary of international research on the subject of "electronic books" that was published between 1993 and 2018 using a particular bibliometric indicator is given. According to the survey, research on "E-books" is expanding quickly in the field of library and information science. The majority (73.25%) of the worldwide research production in the literature is accounted for by just the top 10 most productive nations in the field. The United States of America holds the top spot in this category with 38.42% of all global publications, followed by distant cousins, the other 9 top countries, with contributions ranging from 1.61 to 8.74 percent. The UK has the greatest relative citation index (12.08) and citation effect per paper (1.61). The most productive author was H. Falk, while the most productive organization was Bar-Ilan University in Israel. In e-books, Publishing Research Quarterly is the most widely read journal.

Gupta et al., (2019) emphasis is placed on the quantitative and qualitative description of international research in "electronic journals (e-journal)" using a number of metrics, including relative citation index, activity index, citations per paper, annual growth, global publications share, and international collaboration works. Data from the SCOPUS database, which covers the years 1990–2017, were used in the study. Global research on the topic produced just 26 highly referenced articles throughout the period, had a low-level citation rate of 5.28 citations per manuscript, and saw a rapid surge in publications of 18.46%. The features of highly cited works in the field are also described in the paper. According to the report, the United States leads the world in global publications share (45.28%) for the subject, followed by the United Kingdom (12.18%), India (5.49%), and other countries. Second, there is a very

dispersed distribution of research among the participating organizations. For example, the top 20 businesses in the field only made-up 16.58 percent of all global publications at that time. The study comes to the conclusion that, in terms of both number and quality of research, "electronic journals" are still in their infancy as a field of study.

Jayasree and Baby (2019) empirical examination of the purpose of this paper is to talk about the importance of e-resources for scientometric research. A collection of scientometric analysis tools is provided. Information gathered from software websites and a search of the literature. There is discussion of citation tracking systems such as CiteseerX, Web of Science, Scopus, and Google Scholar citations. Bibexcel, CiteSpace, Histcite, Pajek, Publish or Perish, Scholarometer, VOS viewer—a tool for creating and visualizing bibliometric networks—CitNet Explorer—a tool for visualizing and analyzing citation networks of publications, and other software tools for bibliometric analysis are discussed. The study comes to the conclusion that a variety of software tools can be combined to perform a comprehensive scientometric analysis, and that researchers and students who are familiar with bibliometric software will be better able to conduct scientometric research.

Bapte and Gedam (2018) elaborate the report shows the Sant Gadge Baba Amravati University's (SGBAU) scientometric profile from 1996 to 2017. Through affiliation-based search, the necessary research output for the study has been extracted from the SCOPUS database. There are 1130 papers total in the research output, with an average of 10.65% citations per paper. The period from 2007 to 2017 saw the publication of 83.98% of articles, which is a very productive block when compared to the 16.02 percent of research output from 1996 to 2006. In 2009, the H-index (17) reached its maximum. International cooperation created 20.08 percent of the documents. The main partnerships are with the United States (21) and Brazil (69) at SGBAU, Amravati. Collaboration trend is predicted by authorship pattern. A maximum of 370 papers were published by two authors; however, the greatest number of citations (4444) for research papers produced by three authors was obtained for their 315 works. The pattern of collaborative authorship has the highest number of citations (89.17%). The average adjusted collaborative coefficient is 0.6289, whereas the average degree of collaboration is 0.96. The most common

source types favored by the SGBAU faculty for research expression are 839 journal and 174 conference papers. The most prolific writers are Mahendra Rai (209), Anand S. Aswar (94), and S. K. Omnwar (143). Alka P. Yadav's 3073 citations for her 17 works are impressive. The field that publishes the majority of articles is Physics and Astronomy (356), which is followed by Chemistry, Material Science, Computer Science, Engineering, and Biochemistry-Genetics-Molecular Biology. The majority of the time, SGBAU faculty members have limited their institutional engagement to the state of Maharashtra. The frequency and co-occurrence of keywords indicate the main topics of research that faculty members are working on within a variety of subject areas. With VOSviewer, data visualization is accomplished.

Bhui and Sahu (2018) reported This study assesses the number of citations received by academic members from the Indian Institute of Technology (IIT) Kharagpur's Humanities and Social Science (HSS) Departments for articles they have published in journals. The publication data is taken from the yearly reports and individual biosketches that are accessible on the websites during the 17-year period between 2000 and 2016. The citation counts were calculated using the Web of Science database. The study looks at parameters like the total number of publications in journals, the most popular journals for publication, highly referenced journals, the collaborative nature of faculty articles and the number of citations they receive, and the faculty member with the highest number of citations. The results show that 359 journals include a total of 623 articles. Over time, there has been a growing tendency in the quantity of publications. The top-ranked journal was Psychological Studies, with 27 faculty publications; nevertheless, the top-ranked journal was Journal of Business Ethics, with the most citations. Three-author collaborative studies got the most citations. Although the faculty members work in a variety of subjects, their influence is greater domestically than globally. It is discovered that Sage is the most favored publisher for academic members from IIT Kharagpur's HSS Departments to publish their journal papers.

Chauhan (2018) designed to examine the growing number of cases of plagiarism occurring worldwide. Plagiarism and other problems also affect India. Numerous academics and researchers have expressed alarm about the quickly increasing number of plagiarism accusations. The average growth of publications on plagiarism by

Indian writers, the level of author collaboration, and the most influential sources for writing on plagiarism were all analyzed in this study. A search on the term "Plagiarism" was used to retrieve data from Scopus for the years 2002 through 2016. 385 publications in all were located and analyzed. In comparison to the two other block periods of the study, the average citations per publication for the period 2007–11 was found to be high at 4.01, whereas the block period 2002–06 experienced an annual growth rate of 49.53 percent. The number of articles with a single authorship was higher, at 34.55 percent. Interestingly, these publications also attained a 50.38 percent citation rate. Additionally, it was noted that with each year that goes by, the level of multi-authored collaboration has increased.

Dutt and Kumar (2018) conducted a Analyze 718 research articles on dye-sensitized solar cells (DSSCs) that were published by Indian scientists between 2011 and 2015 and were listed in the Web of Science index. It examined the whole spectrum of research performance by institutions, performing sectors, co-authorship, citations, and various collaboration elements. Private universities are among the best-performing establishments, and DSSC research was dominated by topics connected to nanotechnology. Since nearly 90% of the DSSC research was published in international journals with a respectable impact factor and was published in the USA, the UK, and other developed European countries, it was determined that the research conducted by Indian academics was in line with the mainstream science in the field.

Gupta et al., (2018) looks at 13021 Indian papers that were included by the Scopus database between 2007 and 2016 on mobile research. The Indian articles were retrieved through a search technique that limits the search to India in the "country tag" and the years 2007–2016 in the "date range tag." The search phrase "mobile" is used in the two tags, "keyword tag" and "Article Title tag." During 2007–16, the Indian mobile research output saw an average annual growth rate of 22.64%, a global share of 5.88%, a citation impact of 2.11, and an international collaborative publication share of 11.28%. Subjects that provided the most to publications between 2007 and 2016 were computer science (74.61%), engineering (38.32%), mathematics and social sciences (6.94% and 6.54), etc. Mobile networks contributed the highest publishing share (53.19%) of the 14 subfields that were discovered in India's mobile research. These were followed by mobile telecommunication systems (35.09%), mobile

communication (29.22%), mobile ad hoc networks (26.86%), mobile security (20.62%), and so on. Together, the top 20 Indian organizations and authors contributed 25.64 percent and 5.11 percent, respectively, to India's total production of publications in mobile research from 2007 to 2016. During 2007–16, the top 20 journals accounted for 23.0% of all Indian journals published. The total number of citations received by the 14 papers ranging from 109 to 548 is 3259, meaning that each article earned an average of 232.79 citations. Draw the conclusion that, in terms of both the number and quality of research conducted on mobile devices, India is still not a global leader in this field. The Indian government needs to designate this field as one of the national priority areas, involving much larger R&D investments and trained manpower in addition to increasing international collaboration with top mobile research hubs, given its strategic and global importance and the need to improve the output and quality of research.

Gupta and Dhawan (2018a) gives a quantitative and qualitative account of research on three-dimensional (3D) printing based on the 7309 papers that have been published worldwide in the subject and are indexed in the Scopus database during the course of a ten-year period, from 2007 to 2016. Over the course of ten years, 3D printing research averaged 10.59 citations per paper and grew at a rate of 54.61% annually. The majority of the world's production in this sector, accounting for 93.79% of all publications, comes from just 16 countries. The papers also shed light on the qualitative performance of 3D printing research in relation to the field's most popular subjects, materials, technologies, and applications, top 25 global organizations and authors, citations per paper, highly cited papers, and relative citation index. According to the study's findings, China and the United States will spearhead the next industrial production revolution—3D printing. Asian and Pacific nations ought to take the lead in promoting 3D printing research by means of suitable financial and policy frameworks, as well as by encouraging research groups to work together with preeminent global centers for 3D printing research.

Gupta and Dhawan (2018b) This report examines the global production of artificial intelligence research, spanning the years 2007 to 2016, with a total of 1,52,655 papers viewed from the Scopus database. The global publication share of artificial intelligence research was 74.32 percent, with the top 10 countries making up this

proportion. Their respective global shares ranged from 3.68% to 19.46%, with China holding the largest share (19.46%), followed by the United States (17.96%), India (6.37%), and the United Kingdom (6.33%), among others. The study on artificial intelligence publications produced in India is also examined in this report. Over the course of ten years, from 2007 to 2016, India amassed 9730 publications overall, with an average annual growth rate of 27.35%. The average citation impact per paper was 2.76, and the country's share of international collaborative publications (CP) during this time was 10.34%. The field with the highest percentage of publications (86.99%) was computer science, followed by engineering (30.69%), mathematics (15.95%), biochemistry, genetics & molecular biology (4.66%), and a few other fields. Between 2007 and 2016, the top 10 organizations and writers collectively accounted for 19.31% and 2.71% of the national publications share and 29.78% and 6.85 percent of the national citation share, respectively. The top ten journals published 1650 papers, or 15.45% of the total output published in journal format across the nation. 24 highly referenced papers, with an average of 162.46 citations per paper, came from India. 109 writers from 70 organizations contributed to these 24 highly referenced works, which were published in 15 journals.

Rochim et al., (2018) observed that one of the bibliometric assessment techniques for researchers' performance has been a widespread use of the H-index. Conversely, H-index has proven to be unjust in identifying authors who are perfectionist researchers (many citations but few papers) and productive researchers (many papers but little citations). This article's major goal is to enhance the H-index by using Lotka's Law and Jain's Fairness Index method to calculate the influence of productive and perfectionist researchers. It is suggested to use RA-index to improve H-index. 1,710 citation data sets of the most highly cited scholars from Scopus—based on an author names list from the Webometrics website—are utilized to support the suggested methodology. The average fairness index of the RA-index is 91%, which is greater than the fairness of the H-Index, which is found to be 80%.

Sadik Batcha (2018) looked over the Currently, dengue is thought to be the most significant virus spread by mosquitoes worldwide. It is challenging to determine the precise impact of dengue worldwide because of things like poor disease surveillance, incorrect diagnoses, and low reporting rates. The social, economic, and disease impact

is probably greatly underestimated by the data that is now available. Dengue fever has been estimated to affect between 50 million and 200 million people annually worldwide. In the last eight years, scientists have published 6,031 research papers totaling 22,234 authors in 6007 scientific journals, reflecting the significant attention this fever has received. The goal of this research is to determine whether Lotka's Law applies to international publications on dengue. In order to assess authors' scientific production, determine Lotka's Exponent value, and determine whether Lotka's Law is fit for use, the study focuses on Lotka's empirical law of scientific productivity, also known as the Inverse Square Law.

2.2 SUMMARY OF THE REVIEW OF THE LITERATURE

The extensive literature survey was conducted to examine the relevance of the literature on Coronavirus and the significance of scientometric analysis. It has been observed that scientometric studies have gained a lot of popularity and its one of the most imperative tools to examine the growth of any field and to evaluate the parameters such as authors' productivity, citation, institute productivity, popular sources and keywords, many other important indicators. This type of analysis enables the researcher to identify the research gap in the previous studies. Scientometric studies help researchers for continuing research, policies making, fund allocation, geographical distribution on particular research area. **Danesh and GhaviDel (2020)** the statistical population of this article includes 5128 Coronavirus subject area documents indexed on the WoS from 1970 to 2019. The keywords were extracted from MeSH and analyzed using Excel 2016. Data analysis showed that the highest science production was in 2005, and the highest citation number was in 2019. "Enjuanes L." is the most proliferated author, the United States, the most productive country, and the University of Hong Kong, the top organization in Coronavirus in the last half-century. (Singh, 2020) As per WHO (4 April 2020) 1,056,159 Confirmed cases and 57,206 deaths from 208 countries have been reported. The top two authors of the field are Mahase E and Drosten C. The most cited paper is Huang Cl, 2020. The maximum contribution is from China and the top most organization is University of Hong Kong, Homg Kong. **Ram and Nisha (2020)** an analysis of 806 highly cited articles was carried out based on the data retrieved from SCOPUS multidisciplinary database on 'Coronavirus' research published during 1970-2019 and 2020. During the

period a total of 18,116 articles published up to 2019; and 1221 articles published in 2020 till 30 March 2020. The articles with more than 100 citations were taken as highly cited articles in coronavirus research. On this basis, there were 806 articles with more than 100 citations and considered for the data analysis. The highly cited articles were mostly published from the USA. The 'Journal of Virology' was most productive journal publishing most of the highly cited articles. These highly cited articles were published between 1973 to 2016. **Hussain et al., (2021)** the study's main purpose was to map Cerebral Palsy (CP), a bibliometric analysis of the highly cited research output, 1990-2020. The web of science database was used to retrieve all records related to CP. The Impact Factor of the reviews has been taken from the 2019 issue of Clarivate Analytics' Journal Citation Report (JCR), named IF2019. The Hirsch Index (h-Index) has been compiled from the most active writers and organizations database. The co-citation networks between efficient authors of highly cited papers were used by Biblioshiny, ScientoPy, and VOS viewer.

2.3 RESEARCH GAPS

After reading thoroughly some best research work conducted on the different fields of Bibliometric and Scientometric, the researcher identifies a number of opportunities on which he may conduct his research. After reading a number of previous studies, he is now enabled to structure his own research work.

It has been found that a number of studies are done in this relevant area, but still some gaps exist in these studies. Some of these gaps are:

- A number of studies have been carried out in the area of Bibliometrics and Scientometric, but very few researches has been carried out on the Coronavirus related research, particularly on COVID-19 and other relevant area.
- As per in my best knowledge that no such effort had been made to analyze the growth of Coronavirus research literature by applying Scientometric techniques.
- To the best knowledge the researcher could not find any type of research on Coronavirus mapping of research output.

Therefore, topic enlisted a Scientometric study of research output on coronavirus has been selected for in study purpose.

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